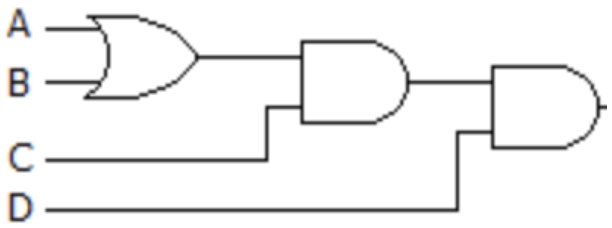
Tutorial 04 - Boolean Functions

1. Draw the truth table for the Boolean function W'(X+Y) Z. Add two more columns and write Minterm and Maxterm expressions for each input combination.

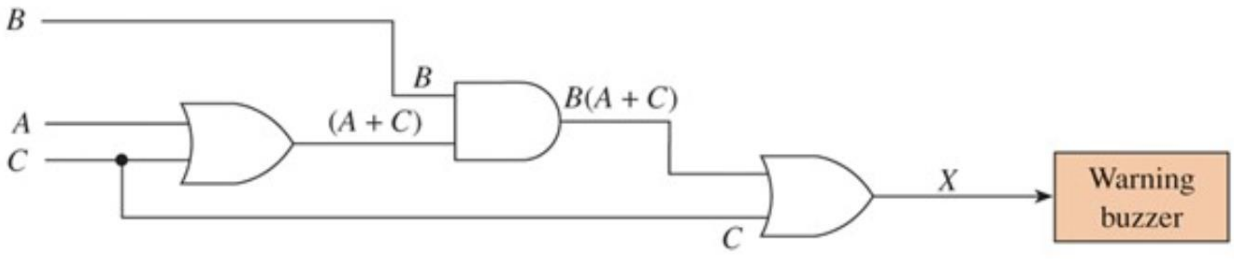
2. Answer the following questions based on the logic circuit shown below.

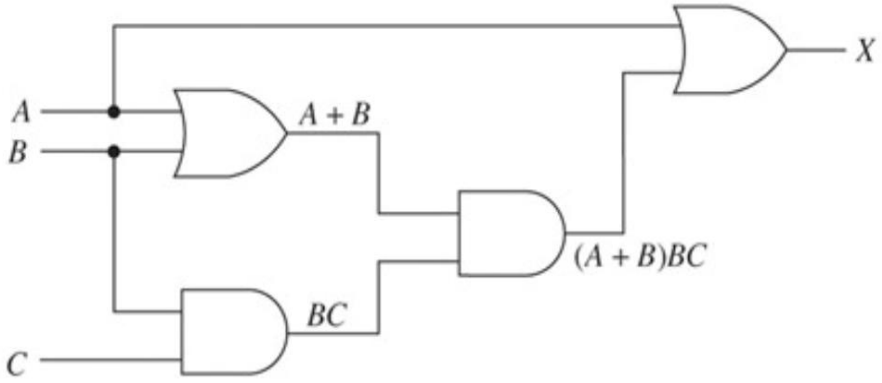


   i. Derive the Boolean expression for the above circuit.
   ii. Draw the truth table for the Boolean expression identified.
   iii. Find Minterm and Maxterm for each input combination.
   iv. Express the Boolean function in SOP and POS forms.

3. Write the Boolean logic equation, and draw the logic circuit and truth table that represents the following scenario: A bank burglar alarm is to activate if it is after banking hours and the front door is opened or if it is after banking hours and the vault door is opened.

4. The logic circuit shown below is used to turn on a warning buzzer at X based on the input connections at A, B, and C. A simplified equivalent circuit that will perform the same function can be formed by using Boolean algebra. Write the equation of the circuit, simplify it and draw the logic circuit of the simplified equation.





5. Inspect each of these Boolean expressions, and determine whether each one is a sum of products (SOP) or a product of sums (POS):

$$(B + \overline{C} + D)(\overline{A} + B)$$

$$A\overline{B}\,\overline{C} + \overline{A}BC$$

$$(X + \overline{Y} + \overline{Z})(\overline{Y} + Z)(\overline{X} + Y)$$

$$\overline{M}\,\overline{N}\,\overline{O} + MN\overline{O} + M\overline{N}O$$

$$(X + \overline{Y + Z})(\overline{Y + \overline{Z}})$$

$$\overline{ABC} + A\overline{B}C$$

6. From the truth table below, determine the standard SOP and POS expression.

Inputs			Output
A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

7. Examine this truth table and then write both SOP and POS Boolean expressions describing the Output:

A	B	C	Output
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

8. Write a Boolean SOP expression for this truth table, then simplify that expression as much as possible, and draw a logic gate circuit equivalent to that simplified expression.

A	B	C	Output
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0